

Table A.1

Descriptive statistics of the model variables for rust

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	348.79	304.00	270.09	(346.85, 350.74)
<i>ns</i>	1.03	1.00	0.21	(1.03, 1.04)
<i>nf</i>	6.43	2.00	92.97	(5.77, 7.11)
<i>rexp</i>	74.79	1.13	1321.50	(65.27, 84.32)
<i>sexp</i>	3835.32	673.5	6940.66	(3785.27, 3885.37)
<i>entropy</i>	0.8812	0.2422	1.2184	(0.87, 0.89)
<i>days_to_first_fix</i>	3.46	0.00	47.83	(3.12, 3.81)
<i>exp</i>	4255.27	889.25	7357.31	(4202.22, 4308.33)

Table A.2

Descriptive statistics of the model variables for Tensorflow

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	268.72	285.00	140.67	(267.65, 269.81)
<i>ns</i>	1.07	1.00	0.36	(1.07, 1.08)
<i>nf</i>	5.23	2.00	35.58	(4.96, 5.51)
<i>rexp</i>	17.61	1.13	246.65	(15.72, 19.51)
<i>sexp</i>	20180.28	596.5	34515.36	(19915.01, 20445.56)
<i>entropy</i>	0.8850	0.6665	1.0911	(0.88, 0.89)
<i>days_to_first_fix</i>	24.60	0.00	87.73	(23.93, 25.28)
<i>exp</i>	21150.18	780.75	35330.65	(20878.64, 21421.73)

A. Dataset Description

A heat map is a kind of data visualization where different data intensities within a dataset are represented by color gradients. It is frequently used to find correlations, patterns, and trends in data, which makes it easier to quickly display complex information. It conveys potential relationships between the variables shown on the x- and y-axes by using different colors, or varying shades of the same color, to indicate various values. Darker hues or shades typically indicate larger or more quantities of the values shown in the heat map. The reddish color represents the positive correlation, and the bluish color represents the negative correlation among the variables. The distribution shapes, variability, and central tendencies will also be ascertained using descriptive statistics. It also enable us to identify outliers and unusual values. In the below sections, we have visualized each software projects using correlation metrics and descriptive statistics.

A.1. Rust

The heat map of the correlation coefficient among the variables of our interest for the Rust data set is shown in Fig. A.1.

Table A.1 shows the statistics that can help to understand the distribution and variability of the different variables in the Rust dataset.

A.2. TensorFlow

The heat map of the correlation coefficient among the variables of our interest for the Tensorflow data set is shown in Fig. A.2.

Table A.2 shows the statistics that can help to understand the distribution and variability of the different variables in the Tensorflow dataset.

A.3. Rails

The heat map of the correlation coefficient among the variables of our interest for the Rails data set is shown in Fig. A.3.

Table A.3 shows the statistics that can help to understand the distribution and variability of the different variables in the Rails dataset.

Table A.3

Descriptive statistics of the model variables for Rails

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	252.61	232.00	169.90	(251.23, 254.00)
<i>ns</i>	1.15	1.00	0.66	(1.15, 1.16)
<i>nf</i>	2.76	1.00	8.20	(2.70, 2.84)
<i>rexp</i>	65.09	1.119	549.86	(60.62, 69.57)
<i>sexp</i>	428.39	77.0	746.29	(422.32, 434.48)
<i>entropy</i>	0.6316	0.00	0.8885	(0.63, 0.64)
<i>days_to_first_fix</i>	106.44	0.00	352.34	(103.57, 109.31)
<i>exp</i>	1713.74	450.0	2613.08	(1692.47, 1735.02)

Table A.4

Descriptive statistics of the model variables for VSCode

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	60.60	63.00	27.40	(60.37, 60.84)
<i>ns</i>	1.05	1.00	0.26	(1.05, 1.05)
<i>nf</i>	4.59	1.00	55.96	(4.11, 5.07)
<i>rexp</i>	52.53	1.40	422.99	(48.89, 56.17)
<i>sexp</i>	4739.20	2312.5	5942.17	(4688.06, 4790.36)
<i>entropy</i>	0.7228	0.00	1.0209	(0.71, 0.73)
<i>days_to_first_fix</i>	2.67	0.00	32.59	(2.40, 2.96)
<i>exp</i>	6601.52	4540.75	6804.46	(6542.96, 6660.10)

Table A.5

Descriptive statistics of the model variables for WP-Calypso

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	138.94	146.00	75.89	(138.11, 139.79)
<i>ns</i>	1.11	1.00	0.39	(1.11, 1.12)
<i>nf</i>	3.85	2.00	27.52	(3.55, 4.16)
<i>rexp</i>	48.91	1.11	337.92	(45.16, 52.66)
<i>sexp</i>	507.74	206.0	778.86	(499.11, 516.39)
<i>entropy</i>	0.77	0.4262	0.99	(0.76, 0.79)
<i>days_to_first_fix</i>	45.27	0.00	154.18	(43.56, 46.98)
<i>exp</i>	701.74	337.0	998.26	(690.67, 712.83)

A.4. VSCode

The heat map of the correlation coefficient among the variables of our interest for the VSCode data set is shown in Fig. A.4.

Table A.4 shows the statistics that can help to understand the distribution and variability of the different variables in the VSCode dataset.

A.5. WP-Calypso

The heat map of the correlation coefficient among the variables of our interest for the WP-Calypso data set is shown in Fig. A.5.

Table A.5 shows the statistics that can help to understand the distribution and variability of the different variables in the WP-Calypso dataset.

A.6. Camel

The heat map of the correlation coefficient among the variables of our interest for the Camel data set is shown in Fig. A.6.

Table A.6 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A.6

Descriptive statistics of the model variables for Camel

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	47.31	31.00	44.80	(46.81, 47.82)
<i>ns</i>	1.28	1.00	0.83	(1.27, 1.28)
<i>nf</i>	6.77	2.00	41.32	(6.30, 7.24)
<i>rexp</i>	63.45	1.06	682.88	(55.75, 71.15)
<i>sexp</i>	5149.49	844.00	8191.84	(5057.15, 5241.85)
<i>entropy</i>	1.0853	0.8136	1.3254	(1.07, 1.10)
<i>days_to_first_fix</i>	48.74	0.00	225.34	(46.20, 51.28)
<i>exp</i>	15312.83	5702.50	19512.00	(15092.86, 15532.80)

Table A.7

Descriptive statistics of the model variables for CoreFX

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	193.68	188.00	109.81	(192.37, 195.01)
<i>ns</i>	1.13	1.00	0.41	(1.13, 1.14)
<i>nf</i>	11.71	2.00	78.34	(10.77, 12.66)
<i>rexp</i>	25.11	1.07	228.37	(22.37, 27.85)
<i>sexp</i>	3538.41	440.00	8407.69	(3437.42, 3639.41)
<i>entropy</i>	1.17	0.83	1.52	(1.16, 1.19)
<i>days_to_first_fix</i>	13.53	0.00	85.95	((12.50, 14.57)
<i>exp</i>	4613.17	788.00	11077.52	(4480.12, 4746.24)

Table A.8

Descriptive statistics of the model variables for Django

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	77.25	35.00	81.19	(76.27, 78.23)
<i>ns</i>	1.45	1.00	0.72	(1.45, 1.46)
<i>nf</i>	4.37	2.00	33.82	(3.97, 4.79)
<i>rexp</i>	41.75	1.03	414.70	(36.74, 46.76)
<i>sexp</i>	802.20	268.00	1432.75	(784.91, 819.50)
<i>entropy</i>	0.7102	0.00	1.0183	(0.70, 0.72)
<i>days_to_first_fix</i>	153.80	0.00	377.60	(149.25, 158.36)
<i>exp</i>	2335.92	836.25	3360.53	(2295.35, 2376.49)

A.7. CoreFX

The heat map of the correlation coefficients among the variables of interest in the CoreFX dataset is shown in Fig. A.7.

Table A.7 shows the statistics that can help to understand the distribution and variability of the different variables in the CoreFX dataset.

A.8. Django

The heat map of the correlation coefficients among the variables of interest in the Django dataset is shown in Fig. A.8.

Table A.8 shows the statistics that can help to understand the distribution and variability of the different variables in the Django dataset.

A.9. Nova

The heat map of the correlation coefficients among the variables of interest in the Nova dataset is shown in Fig. A.9.

Table A.9 shows the statistics that can help to understand the distribution and variability of the different variables in the Nova dataset.

Table A.9

Descriptive statistics of the model variables for Nova

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	168.82	149.00	118.28	(167.40, 170.25)
<i>ns</i>	1.14	1.00	0.44	(1.14, 1.15)
<i>nf</i>	4.49	2.00	36.56	(4.05, 4.94)
<i>rexp</i>	124.65	1.04	2619.02	(93.01, 156.30)
<i>sexp</i>	363.62	90.0	744.30	(354.63, 372.62)
<i>entropy</i>	0.8764	0.7219	1.0609	(0.86, 0.89)
<i>days_to_first_fix</i>	92.37	0.00	220.29	(89.71, 95.04)
<i>exp</i>	532.47	164.5	980.44	(520.62, 544.32)

Table A.10

Descriptive statistics of the model variables for JGroup

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	10.83	11.00	7.07	(10.73, 10.94)
<i>ns</i>	1.17	1.00	0.48	(1.17, 1.18)
<i>nf</i>	2.51	1.00	13.09	(2.32, 2.70)
<i>rexp</i>	47.87	1.0039	1232.16	(29.88, 65.88)
<i>sexp</i>	3342.80	2539.0	3167.17	(3296.54, 3389.07)
<i>entropy</i>	0.3983	0.0	0.8194	(0.39, 0.41)
<i>days_to_first_fix</i>	73.14	0.00	312.84	(68.58, 77.72)
<i>exp</i>	8293.86	7846.0	6150.00	(8204.03, 8383.71)

Table A.11

Descriptive statistics of the model variables for Fabric 8

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	32.10	31.00	16.72	(31.82, 32.40)
<i>ns</i>	1.53	1.00	1.79	(1.51, 1.57)
<i>nf</i>	9.46	2.00	84.56	(7.98, 10.95)
<i>rexp</i>	186.125	1.20	3249.38	(129.14, 243.10)
<i>sexp</i>	895.08	179.00	1469.96	(869.31, 920.86)
<i>entropy</i>	1.0616	0.2108	1.5323	(1.03, 1.09)
<i>days_to_first_fix</i>	25.58	0.00	97.95	(23.86, 27.30)
<i>exp</i>	6918.40	3954.00	8384.15	(6771.39, 7065.43)

A.10. JGroup

The heat map of the correlation coefficients among the variables of interest in the JGroup data set is shown in Fig. A.10.

Table A.10 shows the statistics that can help to understand the distribution and variability of the different variables in the JGroup dataset.

A.11. Fabric 8

The heat map of the correlation coefficient among the variables of our interest for the fabric data set is shown in Fig. A.11.

Table A.11 shows the statistics that can help to understand the distribution and variability of the different variables in the fabric dataset.

A.12. Broadleaf

The heat map of the correlation coefficient among the variables of our interest for the Broadleaf data set is shown in Fig. A.12.

Table A.12 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A.12

Descriptive statistics of the model variables for Broadleaf

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	21.07	24.00	11.00	(20.88, 21.27)
<i>ns</i>	1.33	1.00	0.96	(1.32, 1.35)
<i>nf</i>	9.77	2.00	102.34	(7.97, 11.58)
<i>rexp</i>	129.34	1.09	879.91	(113.81, 144.86)
<i>sexp</i>	1643.81	327.50	3920.65	(1574.62, 1713.01)
<i>entropy</i>	1.0388	0.5586	1.3546	(1.01, 1.06)
<i>days_to_first_fix</i>	48.78	0.00	202.03	(45.21, 52.34)
<i>exp</i>	4628.11	1205.00	7577.76	(4494.37, 4761.85)

Table A.13

Descriptive statistics of the model variables for Brackets

Variable	Mean	Median	SD	Confidence Interval
<i>ndev</i>	42.13	41.00	26.13	(41.66, 42.61)
<i>ns</i>	1.12	1.00	0.34	(1.11, 1.12)
<i>nf</i>	3.05	1.00	26.25	(2.58, 3.53)
<i>rexp</i>	43.76	1.61	264.21	(38.95, 48.57)
<i>sexp</i>	365.39	143.00	511.38	(356.09, 374.71)
<i>days_to_first_fix</i>	58.59	0.00	214.69	(54.69, 62.50)
<i>entropy</i>	0.4887	0.00	0.7824	(0.47, 0.50)
<i>exp</i>	668.61	246.00	1017.29	(650.10, 687.13)

A.13. Brackets

The heat map of the correlation coefficient among the variables of our interest for the Brackets data set is shown in Fig. A.13.

Table A.13 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Appendix

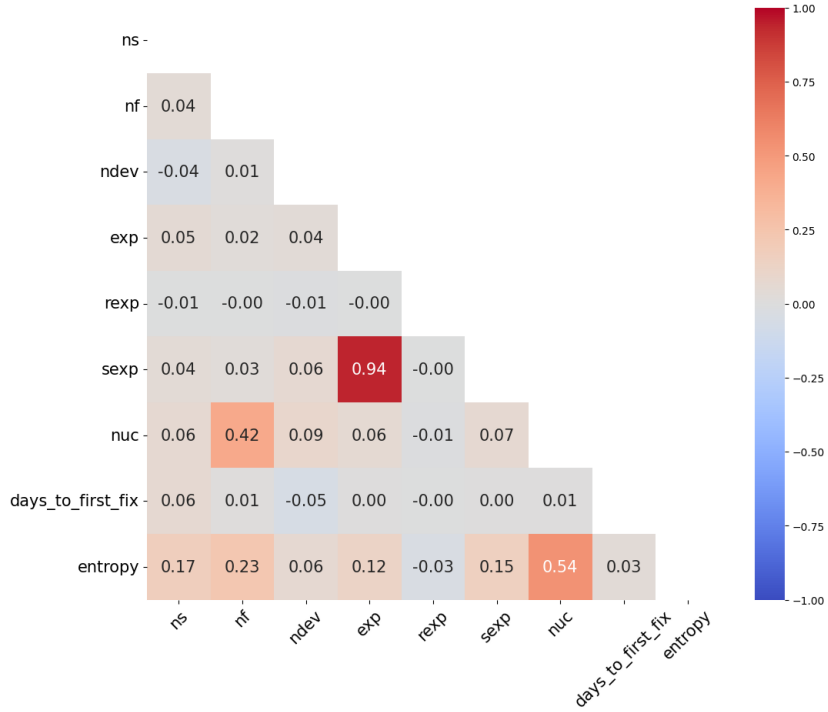


Figure A.1: Heat map of correlation coefficients of Rust dataset

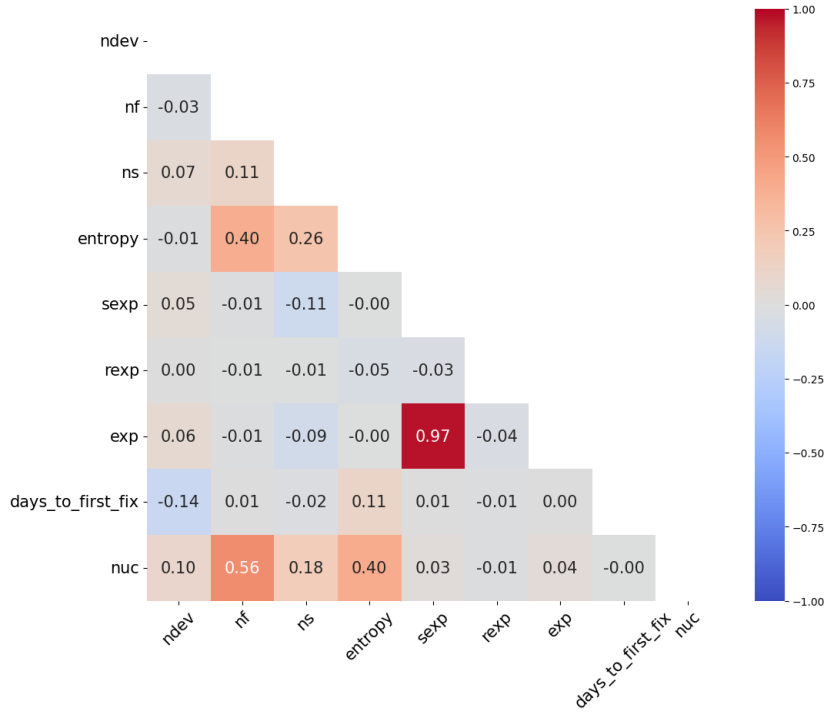


Figure A.2: Heat map of correlation coefficients of Tensorflow dataset

Appendix

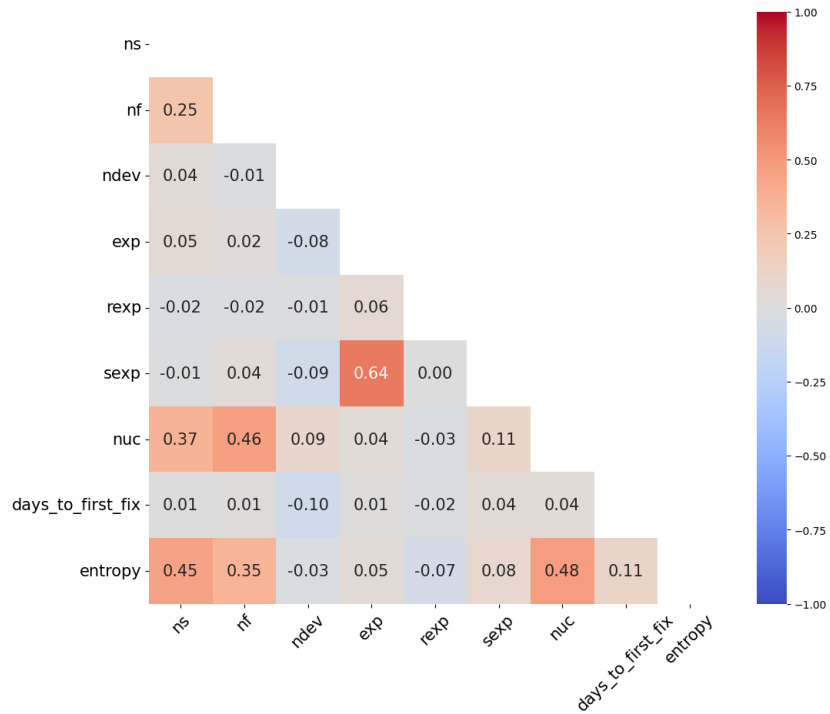


Figure A.3: Heat map of correlation coefficients of Rails dataset

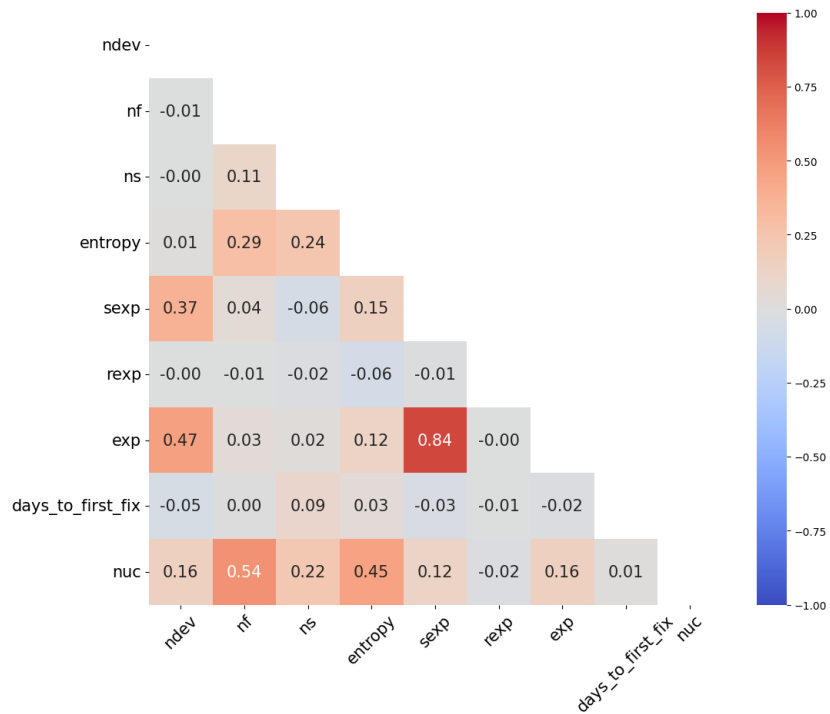


Figure A.4: Heat map of correlation coefficients of VSCode dataset

Appendix

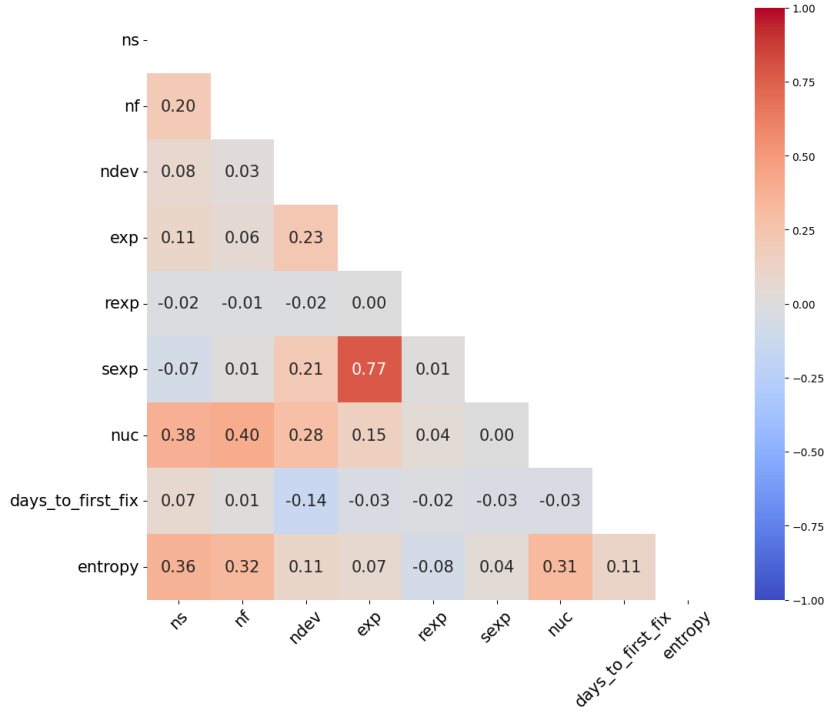


Figure A.5: Heat map of correlation coefficients of WP-Calypso dataset

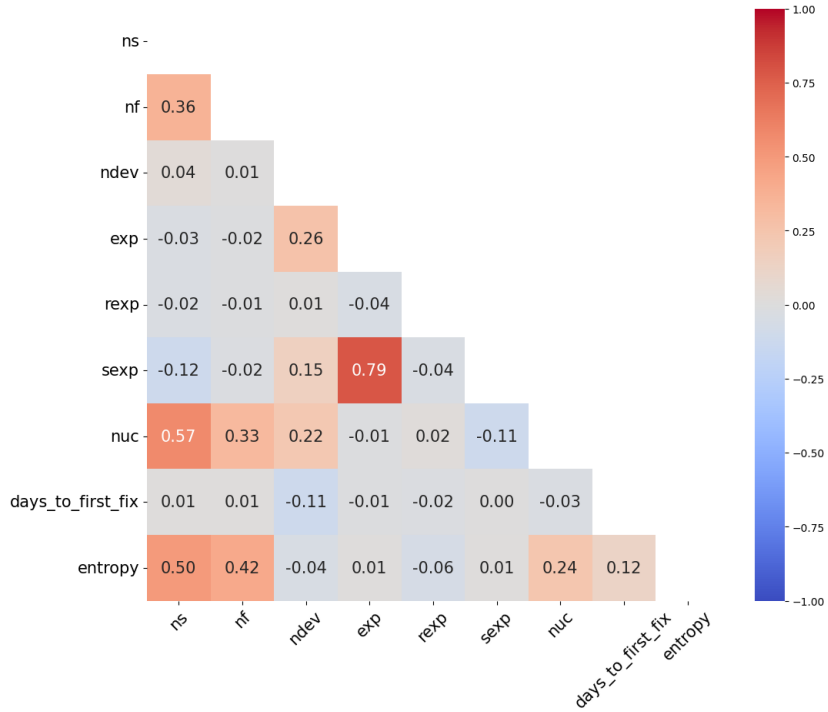


Figure A.6: Heat map of correlation coefficients of Camel dataset

Appendix

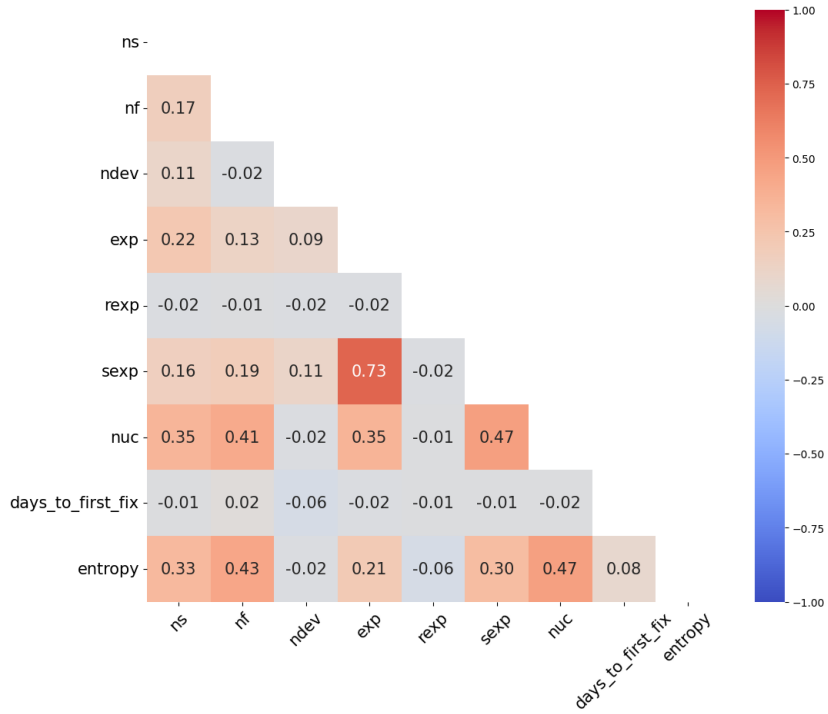


Figure A.7: Heat map of correlation coefficients of CoreFX dataset

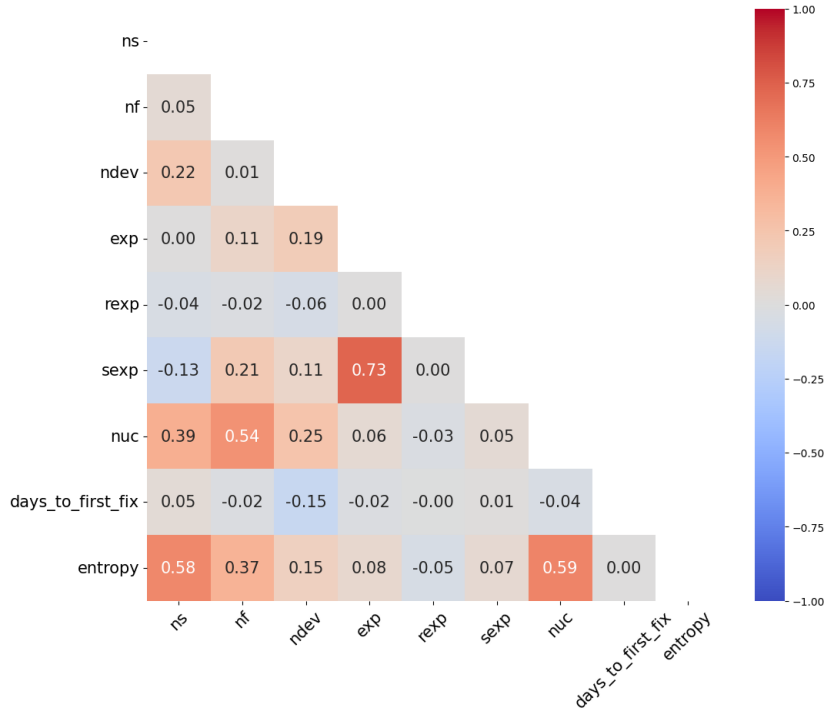


Figure A.8: Heat map of correlation coefficients of Django dataset

Appendix

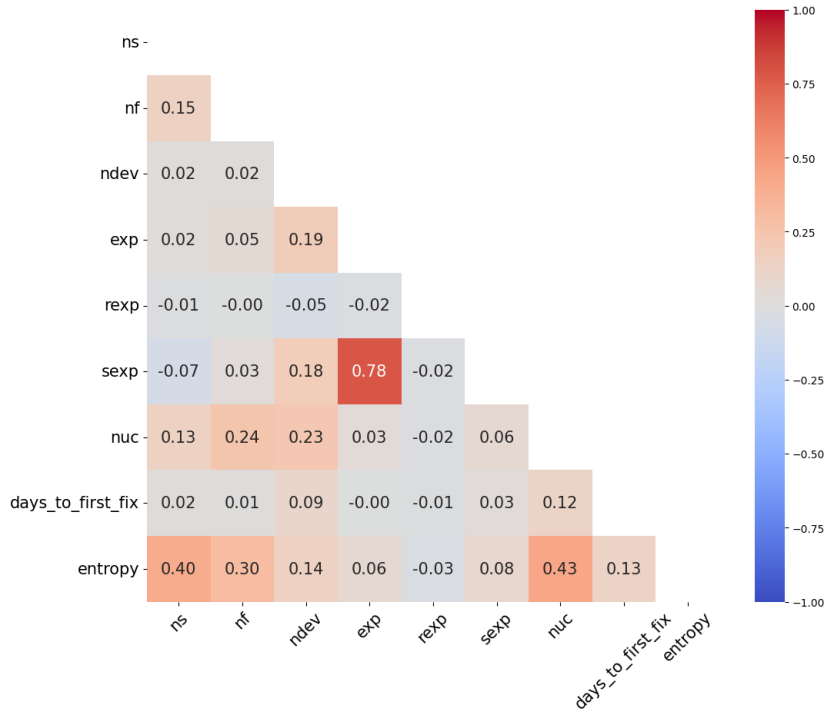


Figure A.9: Heat map of correlation coefficients of Nova dataset

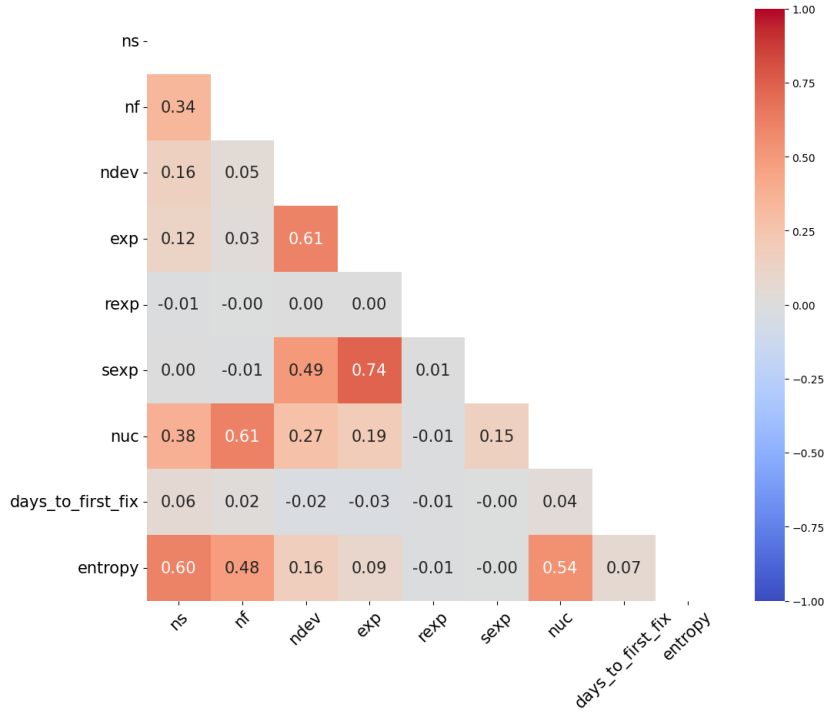


Figure A.10: Heat map of correlation coefficients of JGroup dataset

Appendix

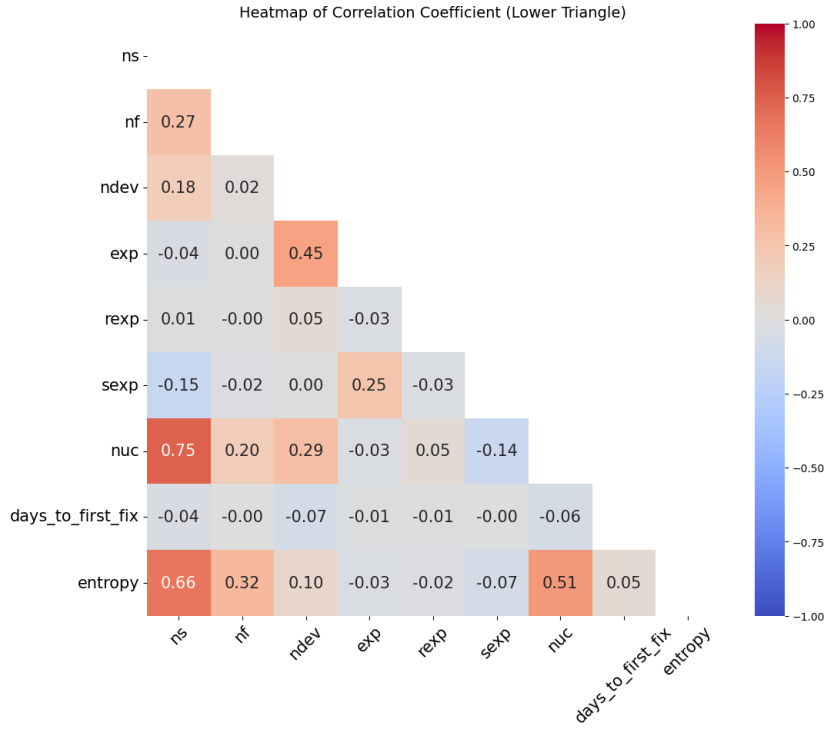


Figure A.11: Heat map of correlation coefficients of Fabric 8 dataset

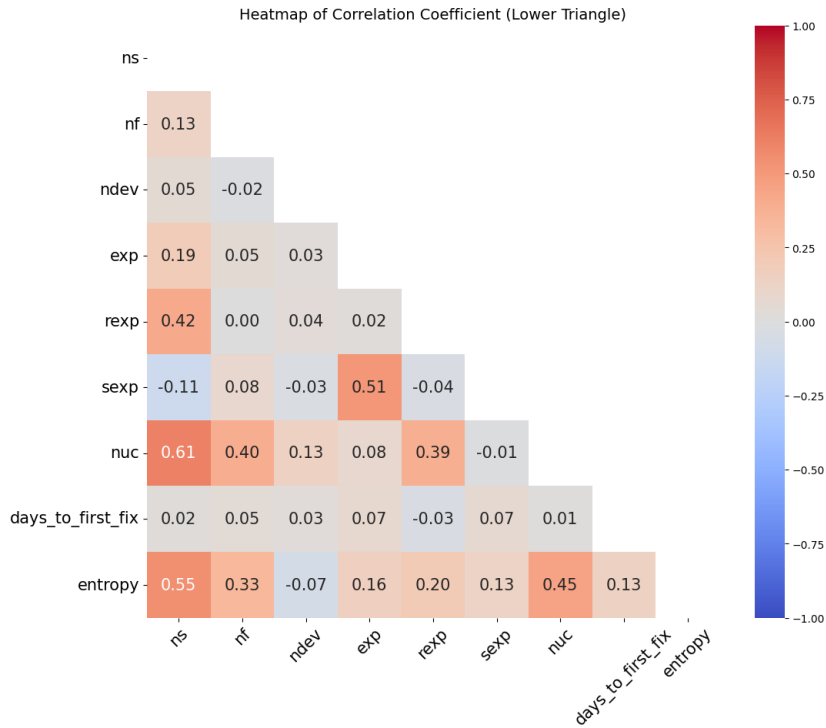


Figure A.12: Heat map of correlation coefficients of Broadleaf dataset

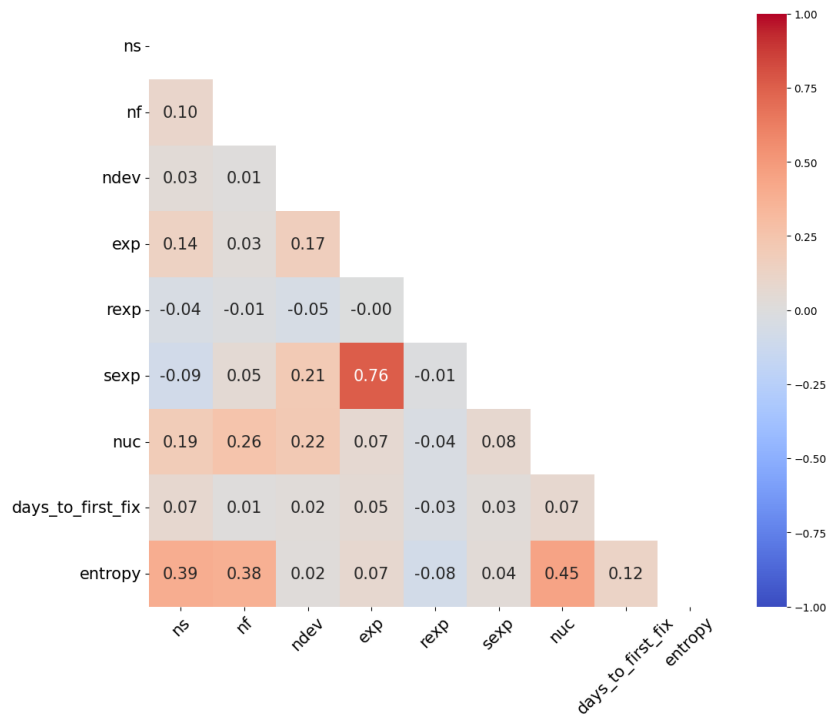


Figure A.13: Heat map of correlation coefficients of Brackets dataset